

KiCad as an Interlocking Plant Design Editor

Purpose: Assess whether KiCad can become the schematic-first editor and source document for FieldUnit Interlocking Plant designs, cTc model boards, fascia/lever boards, and field-device mappings.

Conclusion

KiCad can plausibly become the canonical editor for this work, provided it is used as a real semantic source through custom railroad symbols, properties, and a railroad-specific compiler/linter. It must not be used only as a drawing canvas.

Key constraint: KiCad provides visual connectivity and physical-layout tooling. A railroad compiler must supply conditional Switch topology, route behavior, cTc semantics, and FieldUnit projections.

What KiCad Fits Well

Schematic: Functional Plant Model

A custom KiCad symbol library can represent:

- Switch 783, Switch 795, and Switch 799 with Normal, Reverse, detector, and field-control ports.
- Signal 784 as the functional junction-protecting signal set.
- Directional Signal Masts and Signal Heads.
- Track Circuits such as 1T1, 3T1, and 5T1.
- Plant boundaries, insulated joints, crossings, bridges, industries, and dark-territory interfaces.
- Signal Route behavior symbols that connect a Signal to required Switch positions, required-clear Track Circuits, and an indication ceiling.

KiCad wires and nets become the logical connectivity model. The railroad linter enforces the railroad-specific rules that KiCad ERC does not know.

PCB: Physical Realization and Human-Visible Displays

- **cTc fascia / lever board:** lever, lock-nameplate, CODE-button, maintainer-call, and jewel-lamp footprints.
- **Model board:** static track graphics, LED/indicator footprints, signal and lock glyphs, labels, and modular panel sections.
- **Field node:** connectors, I/O expanders, detector inputs, switch-machine outputs, and signal-head outputs.

Model-board lines do not need electrically meaningful copper. Footprint graphics, silkscreen, fabrication layers, and named LED pads can provide the physical presentation.

Semantic Model in KiCad

Each custom symbol or footprint carries structured railroad properties. Canonical entity names are singular and stable; the cTc machine maps its own columns and nameplates to those entities.

```
Reference: SW795
railroad.kind: switch
railroad.name: 795
railroad.detector: TC3T1
railroad.normal-port: N
railroad.reverse-port: R

Reference: SIG784
railroad.kind: signal
railroad.name: 784
railroad.protects: CP_Luchessa

Reference: CTC-C6
railroad.kind: panel-column
railroad.column: 6
railroad.control: SW795
railroad.nameplate: LOCK
railroad.locked-position: NORMAL
railroad.unlocked-control: local
```

Examples of canonical references: **SW783**, **SW795**, **SW799**, **SIG784**, and **TC1T1**.

Critical Limitation: Conditional Switch Topology

Do not model a Switch as a conventional electrically connected three-pin part where all legs merge into one KiCad net. That would erase the Normal/Reverse path distinction.

- Keep each track leg on a distinct named net.
- Give the Switch symbol named Normal and Reverse ports.
- Let the railroad compiler interpret those ports as a conditional track connection.
- Use KiCad ERC for missing pins and basic connectivity; use the railroad linter for route and safety validation.

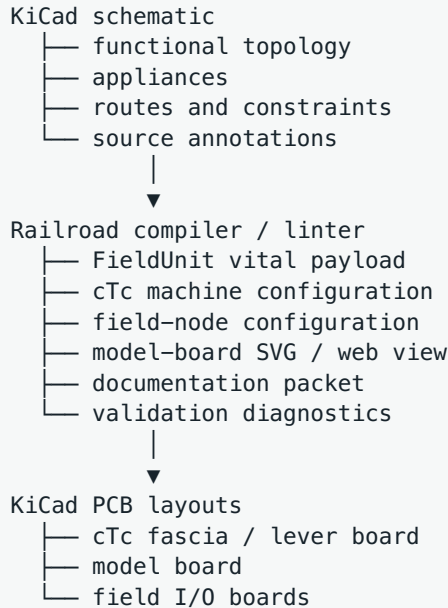
Behaviors as First-Class Schematic Objects

Route behavior must not be free-text documentation or a sidecar JSON file. Use a custom Signal Route symbol/directive with explicit references.

```
Reference: RT-SB-MT-MT2
railroad.kind: signal-route
railroad.signal: SIG784
railroad.mast: SIG784S
railroad.head: SIG784SB
railroad.alignments: SW783=REVERSE,SW795=NORMAL,SW799=NORMAL
railroad.required-clear: TC1T1,TC3T1,TC5T1
railroad.indication: DIVERGING_CLEAR
```

The compiler then derives the FieldUnit vital configuration, cTc bindings, model-board SVG, documentation packet, and later simulation/topology bindings.

Target Architecture



The schematic is the functional source. The PCB is the physical/display realization. Both remain KiCad-native, version-controlled, and human-editable.

Why This Improves the Current Model

- No standalone appliance list plus duplicated panel-column appliance list.
- No standalone route data plus separately maintained route presentation.
- No diagram geometry that can silently drift from functional connectivity.
- No cTc layout detached from the plant model.
- A Switch is a symbol; a route is a behavior symbol; a cTc control is a footprint mapped to that symbol; a model-board lamp is a footprint mapped to an appliance or Track Circuit.

Recommended Tracer: CP Luchessa

1. Create KiCad symbols for Switch, Signal, Signal Mast, Signal Head, Track Circuit, Plant Boundary, and Signal Route.
2. Draw Luchessa from the Graffle/PDF evidence.
3. Add Switches 783, 795, and 799; Signal 784; directional masts/heads; and the reviewed current routes.
4. Create a small cTc PCB/fascia layout for the three Luchessa columns.
5. Write one parser/linter that extracts named objects and emits a FieldUnit vital payload plus a route table.
6. Compare generated documentation with the Graffle and verify the resulting cTc behavior against the Model 503 desk.

If this tracer works, KiCad is a viable editor foundation. If route symbols or conditional topology prove awkward, the experiment identifies the exact small editor capability FieldUnit-Studio must supply.